

DevOps AI Incident Intelligence Assistant

1. Problem Statement

Modern cloud-native applications generate thousands of logs every minute from Linux servers, Docker containers, Kubernetes clusters, and microservices. During incidents, DevOps engineers spend significant time manually reviewing these logs to determine the root cause.

Current monitoring tools such as log viewers and dashboards provide raw logs and metrics but do not explain:

- Why the incident occurred.
- Which log entries are most important.
- How to resolve the issue.
- How similar incidents were solved previously.

Engineers often search documentation, browse online resources, and rely on personal experience, increasing Mean Time To Resolution (MTTR) and causing longer service downtime.

An intelligent AI-powered incident analysis system is needed to automatically understand logs, reason about failures, retrieve relevant troubleshooting knowledge, search the web for recent solutions when necessary, and generate actionable recommendations.

2. Proposed Solution

The **AI DevOps Incident Intelligence Assistant** is an intelligent incident analysis platform that helps DevOps engineers diagnose infrastructure failures quickly using Artificial Intelligence, Retrieval-Augmented Generation (RAG), workflow orchestration, and automation.

Instead of simply displaying logs, the system performs complete incident reasoning.

Step 1 — User Authentication

- Secure Sign Up / Sign In
- JWT Authentication
- User-specific chat history

Step 2 — Log Upload

The engineer uploads:

- Linux logs
- Docker logs

- Kubernetes logs
- Application logs
- JSON logs

The system validates and parses the uploaded file.

Step 3 — Log Parsing

The parser extracts:

- Errors
 - Warnings
 - Failed services
 - Stack traces
 - Exit codes
 - Timestamps
 - Critical events
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Step 4 — AI Workflow (LangGraph)

LangGraph manages the complete reasoning pipeline.

The workflow:

- Parse uploaded log
 - Detect incident type
 - Search knowledge base
 - Perform semantic retrieval
 - Search latest solutions using DuckDuckGo
 - Combine all evidence
 - Generate final incident analysis
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Step 5 — Retrieval-Augmented Generation (RAG)

The system retrieves relevant technical knowledge using Google Gemini Embeddings.

Knowledge includes:

- Kubernetes troubleshooting
- Docker troubleshooting
- Linux administration
- Infrastructure best practices
- Previous troubleshooting documents

Step 6 — Live Web Search

If the local knowledge base is insufficient, the system performs live web search using DuckDuckGo to retrieve recent troubleshooting information.

Step 7 — AI Incident Analysis

Groq Llama 3 analyzes all available information and generates:

- Executive Summary
 - Severity Level
 - Incident Type
 - Confidence Score
 - Root Cause Analysis
 - Supporting Evidence
 - Step-by-step Resolution
 - Prevention Recommendations
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Step 8 — Chat-Based Follow-up

Users can continue asking questions within the same chat session.

Examples:

- Why did this happen?
- Explain this error.
- Give Linux commands.
- How can I prevent this issue?
- Show another solution.

The assistant remembers the conversation using stored chat history.

Step 9 — Automation (n8n)

When a new incident is analyzed:

- Incident summary is automatically emailed using Gmail.
 - The workflow can be extended to create tickets in Jira, GitHub Issues, ServiceNow, or other ITSM platforms.
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Step 10 — History Dashboard

Users can:

- View previous analyses
 - Continue old conversations
 - Review incident summaries
 - Track previous troubleshooting sessions
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3. Technology Stack

Frontend

- React.js
 - Tailwind CSS
 - React Router
 - Axios
 - Framer Motion
 - Lucide React
 - Vercel (Deployment)
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Backend

- FastAPI
 - SQLAlchemy
 - PostgreSQL
 - JWT Authentication
 - Render (Deployment)
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Artificial Intelligence

- Groq (Llama 3)
 - LangGraph
 - Google Gemini Embeddings
 - DuckDuckGo Search API
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Database

- PostgreSQL

Stores:

- Users
- Chat Sessions
- Messages
- AI Analysis
- Incident History

Automation

- n8n

Deployment

Frontend

- Vercel

Backend

- Render

Database

- PostgreSQL

Complete Project Workflow



